

Hydrogen Atom Model

Proton fixed at origin

Electron moving in space

$$q_p = +e, q_e = -e$$

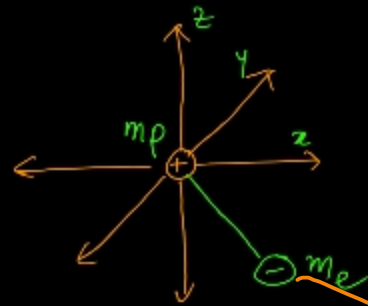
$$e = 1.602 \times 10^{-19} \text{ C}$$

$$m_p = 1.673 \times 10^{-27} \text{ kg} \parallel m_e = 9.109 \times 10^{-31} \text{ kg}$$

$$\frac{m_p}{m_e} = 1823 \quad \mu = \frac{m_e m_p}{m_e + m_p} = 0.9995 m_e$$

$$\hat{H} = \hat{T} + \hat{V}$$

$$\hat{T} = -\frac{\hbar^2}{2m_e} \nabla^2 \quad \nabla^2 \xrightarrow{\text{Laplacian operator}} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$



Coulomb potential

$$V(r) = \frac{q_1 q_2}{4\pi \epsilon_0 r} = \frac{-e^2}{4\pi \epsilon_0 r}$$

permittivity $\epsilon_0 = 8.854 \times 10^{-12} \frac{\text{C}^2}{\text{J} \cdot \text{m}}$

$$\hat{H} = -\frac{\hbar^2}{2m_e} \nabla^2 - \frac{e^2}{4\pi \epsilon_0 r}$$

$$V = V(r) \rightarrow \psi = \psi(r, \theta, \phi)$$

$$\hat{H}\psi(r, \theta, \phi) = E\psi(r, \theta, \phi)$$

$$E_n = -\frac{m_e e^4}{32 \pi^2 \epsilon_0^2 \hbar^2} \left(\frac{1}{n^2} \right) \quad n = 0, 1, 2, \dots$$

n = principle Quantum number

Spherical polar coordinate
Laplacian operator

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

Hydrogen atom energies

$$\hat{H} = -\frac{\hbar^2}{2m_e} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

$$E_n = -\frac{m_e e^4}{32\pi^2 \epsilon_0^2 \hbar^2} \left(\frac{1}{n^2}\right)$$

$n=1, 2, 3, \dots$

principle quantum number.

$g_n = n^2$ degeneracy

$$a_0 = \frac{\epsilon_0 \hbar^2}{\pi m_e e^2} = \frac{4\pi \epsilon_0 \hbar^2}{m_e e^2} = 0.529 \text{ \AA}$$

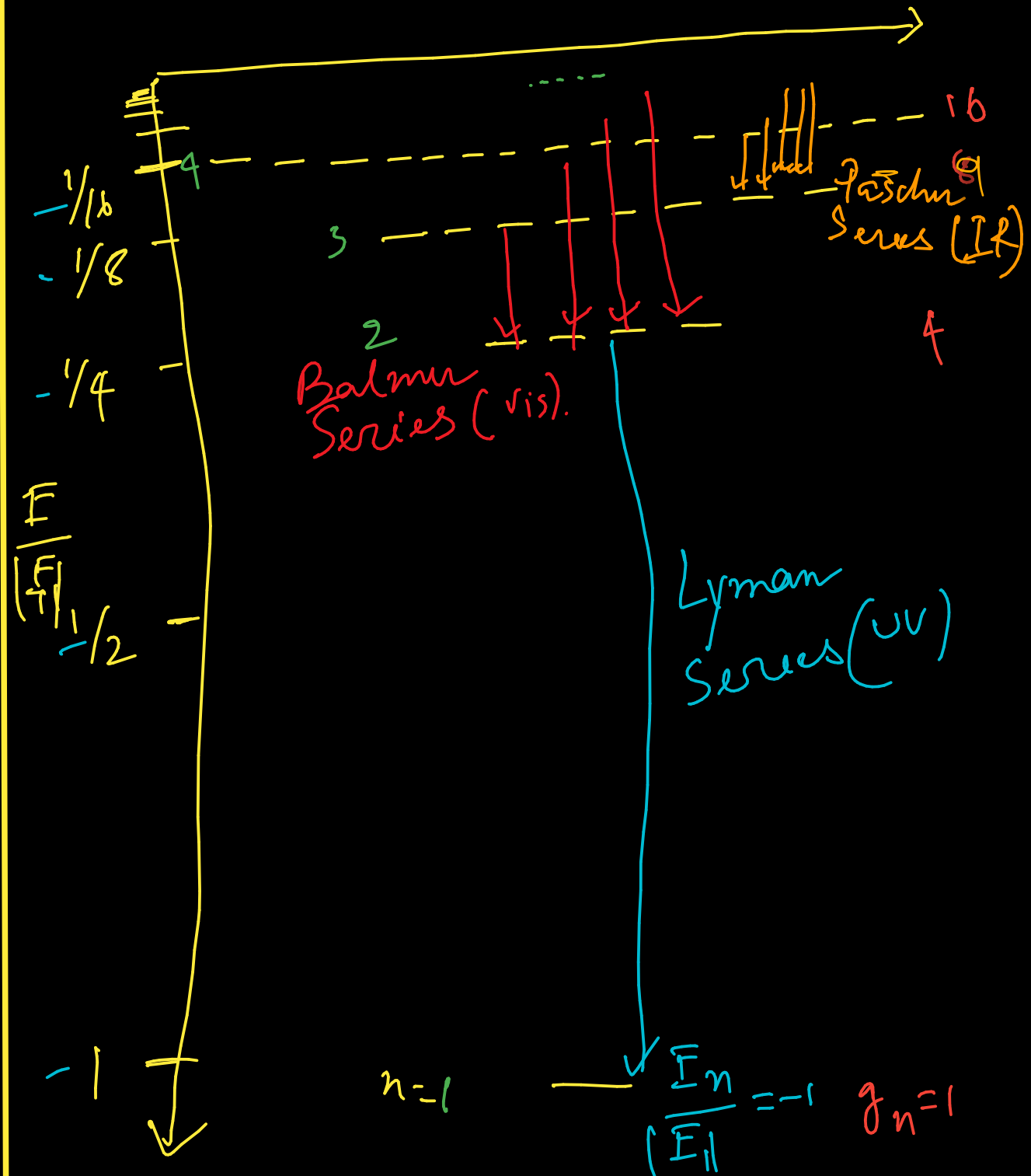
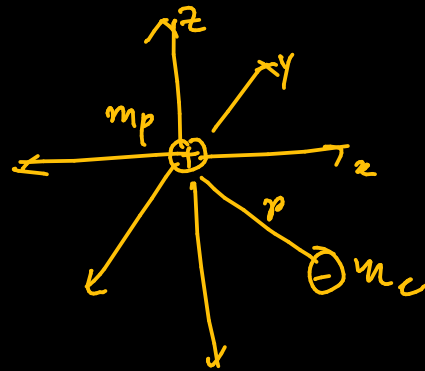
$= 52.9 \text{ pm}$

Bohr radius

$$E_n = -R_H \left(\frac{1}{n^2}\right) \text{ Rydberg formula / Bohr model}$$

$$E_1 = -2.180 \times 10^{-18} \text{ J} = -13.61 \text{ eV} = -1313 \text{ kJ/mol}$$

$$= -313.8 \text{ kcal/mol} = -109,737.3 \text{ cm}^{-1}$$

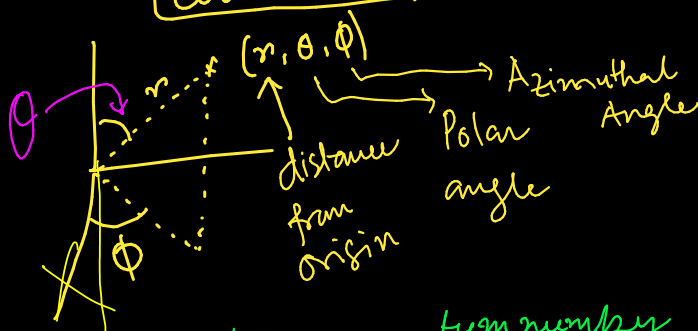


How to find n, l, m ?

H-atom wave function

$$\Psi_{n,l,m}(r, \theta, \phi) = \underbrace{R(r)}_{\text{Radial part (n, l)}} \underbrace{\Theta(\theta) \Phi(\phi)}_{\text{Angular part (l, m)}}$$

Spherical coordinates



n = principle quantum number

l = Angular

m = Azimuthal

SHO wave function, $\Psi_n = \frac{N}{\sqrt{2\pi}} H_n(\xi) e^{-\xi^2/2}$

$$\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}$$

$$\Theta_{lm}(\theta) = \underbrace{C}_{\text{Constant}} \sqrt{\frac{2l+1}{2} \frac{(l-m)!}{(l+m)!}} \underbrace{P_l^m \cos \theta}_{\text{Legendre polynomial}}$$

constant $\downarrow$

$$C = \begin{cases} 1, & \text{when } m \leq 0 \\ (-1)^m, & \text{when } m > 0 \end{cases}$$

Ex, $m=1$
 $C = (-1)^1 = -1$
 $m=2, C = (-1)^2 = 1$

$$P_l^m(x) = (1-x^2)^{|m|/2} \frac{d^{|m|}}{dx^{|m|}} P_l(x)$$

$l=0, m=0 \Rightarrow P_0^0(x) \text{ or } (P_{0,0})$

$$P_0^0(x) = (1-x^2)^{0/2} \frac{d^0}{dx^0} P_0(x)$$

$$P_0^0(x) = 1$$

$l=1, m=0$

$$P_1^0(x) = (1-x^2)^{0/2} \frac{d^0}{dx^0} P_1(x)$$

$$= x$$

Conclusion:

$$P_l^m(x) = P_l(x) \text{ if } m=0$$

if $l=1, m=1 \Rightarrow$

$$P_1^1 = (1-x^2)^{1/2} \frac{d^1}{dx^1} P_1(x) = (1-x^2)^{1/2} \cdot x$$

$$P_1^1 = \sqrt{(1-x^2)} \cdot x$$

Now,

$$\Theta_{0,0}(\theta) = 1 \sqrt{\frac{1}{2} \cdot \frac{1}{1}} P_0^0(x)$$

$$= \frac{1}{\sqrt{2}}$$

Spherical harmonics

$$Y_{l,m}$$

$$= \Theta_{lm}(\theta) \Phi_m(\phi)$$

$$Y_{(0,0)} = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2\pi}} = \frac{1}{2\sqrt{\pi}}$$

Formula:	$P_l^m(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2-1)^l$	$P_0(x) = 1$	$P_0^0(x) = 1$
	$P_0^0(x) = \frac{1}{2^0 0!} \frac{d^0}{dx^0} (x^2-1)^0$	$P_1(x) = x$	$P_1^0(x) = x$
	$= 1$	$P_2(x) = \frac{3x^2-1}{2}$	$P_2^0(x) = \frac{3x^2-1}{2}$
	$P_1^1(x) = \frac{1}{2^1 1!} \frac{d^1}{dx^1} (x^2-1)$		$P_1^1(x) = x$
	$= \frac{1}{2} \times 2x = x$		$P_2^1(x) = \frac{3x^2-1}{2}$
	$P_2^2(x) = \frac{1}{2^2 2!} \frac{d^2}{dx^2} (x^2-1)^2$		$P_2^2(x) = \frac{3x^2-1}{2}$
	$= \frac{1}{8} \frac{d^2}{dx^2} (x^4+1-2x^2)$		$P_3^0(x) = \frac{5x^3-3x}{2}$
	$= \frac{1}{8} \frac{d}{dx} (4x^3+0-4x)$		$P_3^1(x) = \frac{5x^3-3x}{2}$
	$= \frac{1}{8} (12x^2-4)$		$P_3^2(x) = \frac{5x^3-3x}{2}$
	$= \frac{3x^2-1}{2}$		$P_3^3(x) = \frac{5x^3-3x}{2}$

Consider $x = \cos \theta$

How to calculate n, l, m ?

$$l=2, m=0 \Rightarrow$$

$$\Theta_{2,0} = 1$$

$$\Theta_{2,0} = 1 \sqrt{\frac{2 \cdot 2 + 1}{2} \times \frac{2}{2}} P_2^0 \cos(\theta)$$

$$= \sqrt{\frac{5}{2}} \times \frac{3 \cos^2 \theta - 1}{2}$$

$$\Phi_m(\phi) \propto \frac{1}{\sqrt{2\pi}} e^{im\phi}$$

$$\Theta_{lm}(\theta) = \epsilon \sqrt{\frac{2l+1}{2} \frac{(l-m)!}{(l+m)!}} P_l^m \cos \theta$$

constant ϵ $\begin{cases} 1, & \text{when } m \leq 0 \\ (-1)^m, & \text{when } m > 0 \end{cases}$

$P_l^m \cos \theta$ Legendre polynomial

$\epsilon = (-1)^m = -1$
 $m=2, \epsilon = (-1)^2 = 1$

$$P_2^0(x) = \frac{3x^2 - 1}{2}$$

$$P_2^0(\cos \theta) = \frac{3 \cos^2 \theta - 1}{2}$$

$$\Psi = R(r) \Theta(\theta) \Phi(\phi), \quad m, l = ? \quad \leftarrow \text{Question}$$

Quick Methods?

Calculate m from $\phi = \frac{1}{\sqrt{2\pi}} e^{+im\phi}$

If wavefunction have $e^{2i\phi}$, $\boxed{m=2}$

$\otimes e^{-2i\phi}$ $\boxed{m=-2}$

$l = \max^m$ power of sum of $\cos \theta$ and $\sin \theta$

Radial part, $R(r) = e^{-Zr/na_0}$

$$e^{-Zr/2a_0}, \quad n=2$$

n = Principle, l = angular, m = magnetic

$$m \Rightarrow e^{im\phi}$$

$$n \Rightarrow e^{-Zr/na_0}$$

minimum value of power of r , pre-exponential form of n

$l \Rightarrow$ Sum of power of $\sin \theta$ & $\cos \theta$

$\Psi = R \Theta \Phi$

$\downarrow \quad \downarrow \quad \rightarrow m$

$n, l \quad l, m$

① $\Theta(l, m) = \cos^4 \theta + \cos^3 \theta \sin^1 \theta + \sin^2 \theta - \sin \theta + 4$

$l=4$

$\downarrow 4 \quad \downarrow 4 \quad \downarrow 2 \quad \downarrow 1$

② $\Theta(l, m) = \cos^4 \theta + \cos^3 \theta \sin^2 \theta + 4$

$l=5$

$\downarrow 4 \quad \downarrow 5$

③ $\Theta(l, m) = 1 \cos^0 \theta$

$l=0$

④ $\Theta(l, m) = \cos^2 \theta \sin^2 \theta + \cos^5 \theta$

$l=5$

$\downarrow 4 \quad \downarrow 5$

⑤ $R = (r + r^2 - 3r^3) e^{-r/4a_0} \quad n=4, l=1$

⑥ $R = (1-r) e^{-r/4a_0} \quad n=4 \quad l \geq 1$

⑦ $R = r(1-r) e^{-r/2a_0} \quad n=2 \quad l=1$

TABLE 6.5

The complete hydrogenlike atomic wave functions for $n = 1, 2$, and 3 . The quantity Z is the atomic number of the nucleus, and $\sigma = Zr/a_0$, where a_0 is the Bohr radius.

$n = 1,$	$l = 0,$	$m = 0$	$\psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-\sigma}$	$\sigma = \frac{Zr}{a_0}$
$n = 2,$	$l = 0,$	$m = 0$	$\psi_{200} = \frac{1}{\sqrt{32\pi}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \sigma) e^{-\sigma/2}$	$\sigma = \frac{Zr}{a_0}$
	$l = 1,$	$m = 0$	$\psi_{210} = \frac{1}{\sqrt{32\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma e^{-\sigma/2} \cos \theta$	
	$l = 1,$	$m = \pm 1$	$\psi_{21\pm 1} = \frac{1}{\sqrt{64\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma e^{-\sigma/2} \sin \theta e^{\pm i\phi}$	
$n = 3,$	$l = 0,$	$m = 0$	$\psi_{300} = \frac{1}{81\sqrt{3\pi}} \left(\frac{Z}{a_0} \right)^{3/2} (27 - 18\sigma + 2\sigma^2) e^{-\sigma/3}$	
	$l = 1,$	$m = 0$	$\psi_{310} = \frac{1}{81} \left(\frac{2}{\pi} \right)^{1/2} \left(\frac{Z}{a_0} \right)^{3/2} (6\sigma - \sigma^2) e^{-\sigma/3} \cos \theta$	
	$l = 1,$	$m = \pm 1$	$\psi_{31\pm 1} = \frac{1}{81\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} (6\sigma - \sigma^2) e^{-\sigma/3} \sin \theta e^{\pm i\phi}$	
	$l = 2,$	$m = 0$	$\psi_{320} = \frac{1}{81\sqrt{6\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma^2 e^{-\sigma/3} (3 \cos^2 \theta - 1)$	
	$l = 2,$	$m = \pm 1$	$\psi_{32\pm 1} = \frac{1}{81\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma^2 e^{-\sigma/3} \sin \theta \cos \theta e^{\pm i\phi}$	
	$l = 2,$	$m = \pm 2$	$\psi_{32\pm 2} = \frac{1}{162\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma^2 e^{-\sigma/3} \sin^2 \theta e^{\pm 2i\phi}$	

$$\sigma = \frac{Zr}{a_0}$$